Hierarchical Coupling Law for Per-Galaxy a_0 Variation: Evidence from SPARC Rotation Curves

Author Name[1]⋆
[1] *Institution, Address, Country*

Accepted XXX. Received XXX; in original form XXX

ABSTRACT

Using per-galaxy fits to 175 SPARC rotation curves, we find that the MOND acceleration scale a_0 varies systematically across galaxies. Within $N = 45$ published-quality galaxies, a three-axis structural core (gas mass $M_{\rm HI}$, host halo mass $M_{\rm host}$, dynamical coherence MeanRun) explains 44 per cent of a_0 variance in leave-one-out cross-validation. Adding a kinematic coupling channel — the residual flat velocity after removing baryonic structural predictions (VfResid) — raises predictive power to 61 per cent. The signal is regime-dependent, strongest in galaxies with $V_{\rm flat} \geq 120$ km s^{-1}.

External validation on $N = 94$ independent SPARC galaxies using frozen training coefficients confirms that the complete hierarchy replicates: structural core alone fails in all regimes, VfResid dominates by 30–76 percentage points even after improving environmental proxy quality, and all hierarchy checks pass (bootstrap $P > 99.6$ per cent).

Physical interpretation reveals that VfResid encodes genuine halo information (dark halo amplitude haloK is the shared leading driver, $r = 0.60$ internal, 0.51 external) but is not reducible to any catalogue observable: 36–69 per cent of its variance is irreducible and still predicts a_0. After removing haloK, this hidden residual activates with $V_{\rm flat}$ (a_0-predictive correlation rises from $r \sim 0.3$ at low $V_{\rm flat}$ to $r \sim 0.8$ at high $V_{\rm flat}$, slope ratio 2.6:1), ruling out pure observability effects. Hypothesis testing favours dynamical integration — accumulated baryon–halo processing that amplifies in deeper potential wells — over halo response, feedback, or assembly history.

We report structured, regime-dependent a_0 variation with external support and a leading physical interpretation, not a claim of definitive non-universality.

Key words: galaxies: kinematics and dynamics – galaxies: fundamental parameters – dark matter – galaxies: haloes – galaxies: spiral

1 INTRODUCTION

The radial acceleration relation (RAR; McGaugh et al. 2016) connects baryonic and observed accelerations in disc galaxies through a single parameter $a_0 \sim 1.2 \times 10^{-10}$ m s^{-2}, originally identified by Milgrom (1983). Whether a_0 is a universal constant or varies systematically across galaxies remains an open question with implications for both dark matter models (Navarro et al. 1996, 1997) and modified gravity theories (Famaey & McGaugh 2012).

The tight observed scatter of the RAR ($\lesssim 0.13$ dex; Lelli et al. 2017) has been interpreted as evidence for a fundamental acceleration scale. However, several studies have challenged the universality of a_0. Rodrigues et al. (2018) argued from Bayesian model comparison that a universal a_0 is statistically disfavoured, while Li et al. (2018) showed that individual per-galaxy RAR fits reveal galaxy-to-galaxy variation exceeding observational uncertainties. Marra et al. (2020) reinforced this challenge with additional statistical tests. In the dark matter framework, simulations can reproduce the RAR empirically (Ludlow et al. 2017; Navarro et al. 2017), but the predicted scatter depends on baryon–halo coupling physics — adiabatic contraction (Blumenthal et al. 1986), feedback-driven core formation (Di Cintio et al. 2014), and the diversity of halo profiles (Oman et al. 2015; Santos-Santos et al. 2020).

The question we address is not whether a_0 is universal, but what determines its galaxy-to-galaxy variation. Using the SPARC database (Lelli et al. 2016), we perform per-galaxy RAR fits to extract individual a_0 values and ask: which observable galaxy properties predict per-galaxy a_0, what is the physical nature of the predictive signal, and does the relationship transfer to independent galaxies?

We report three main results. First, a hierarchical empirical law in which a structural core (gas mass, halo environment, dynamical coherence) provides the baseline and a kinematic residual channel (VfResid) carries the dominant coupling signal. Second, comprehensive external validation on $N = 94$ independent galaxies confirming that the hierarchy replicates with the same structure, channel ordering, and regime dependence. Third, physical interpretation identifying dynamical integration — accumulated baryon–halo processing — as the leading explanation for the hidden physics inside VfResid.

⋆ E-mail: author@institution.edu

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Table 1. Variable definitions.

Variable	Definition	Source
$\log a_0$	$\log_{10}$(per-galaxy a_0) in $(\mathrm{km\,s^{-1}})^2\,\mathrm{kpc}^{-1}$	RAR fit
$\log M_{\mathrm{HI}}$	$\log_{10}(M_{\mathrm{HI}}/10^9\ \mathrm{M}_\odot)$	SPARC
$\log M_{\mathrm{host}}$	$\log_{10}$(estimated host halo mass)	Tidal analysis
logMR	$\log_{10}$(mean run length in RAR residuals)	RAR fit
$\log V_{\mathrm{flat}}$	$\log_{10}$(asymptotic flat velocity)	SPARC
VfResid	$\log V_{\mathrm{flat}}$ minus structural prediction	Derived
lhOuter	Outer improvement of log-halo over Newton	RC fit
haloK	$\log_{10}$(dark halo linear scale factor)	RC fit

Table 2. Structural core: individual LOO contributions.

Axis	Role	LOO gap (alone)
$\log M_{\mathrm{HI}}$	Gas content	∼25 per cent
$\log M_{\mathrm{host}}$	Host halo environment	∼20 per cent
logMR	Dynamical coherence	∼15 per cent
Combined		44.1 per cent

Table 3. Model hierarchy: LOO gap progression.

Model	LOO gap
Core (3-axis)	44.1 per cent
Core + VfResid (4-axis)	61.1 per cent
Core + VfResid + lhOuter (5-axis)	65.4 per cent

Table 4. Regime dependence of VfResid coupling.

Regime	N	Core+VfResid Δ	Bootstrap P
All galaxies	45	+17.1 pp	99.9 per cent
$V_{\mathrm{flat}} \geq 120\ \mathrm{km\,s^{-1}}$	35	+27.8 pp	99.7 per cent
$V_{\mathrm{flat}} < 120\ \mathrm{km\,s^{-1}}$	10	unstable	18.6 per cent

2 DATA AND METHODS

2.1 Sample

We use the SPARC database (Lelli et al. 2016), comprising 175 disc galaxies with *Spitzer* 3.6 μm photometry and high-quality H I/Hα rotation curves. Per-galaxy a_0 values are obtained by fitting the McGaugh et al. (2016) RAR formula,

$$g_{\mathrm{obs}} = \frac{g_{\mathrm{bar}}}{1 - \exp\left(-\sqrt{g_{\mathrm{bar}}/a_0}\right)}, \tag{1}$$

to each galaxy's rotation curve using marginalized stellar mass-to-light ratios (Υ_{disc} prior: $\mathcal{N}(0.50, 0.12^2)$, range $[0.20, 0.80]$).

The primary analysis sample ('Stage A') consists of 56 galaxies with complete derived properties, of which 45 are classified as published quality and 11 as lower quality. The external validation sample consists of 94 SPARC galaxies not included in Stage A: the original 59 galaxies from **(alias?)** with sufficient RAR data points, plus 35 additional galaxies recovered via sample salvage (Section 3.8).

2.2 Variables

Table 1 defines the key variables used throughout.

2.3 Statistical framework

All predictive claims use leave-one-out (LOO) cross-validation (Stone 1974). The gap percentage is defined as $100 \times (1 - \mathrm{RMSE}^2_{\mathrm{LOO}}/\mathrm{SD}^2)$, measuring variance explained out of sample. For transfer validation, the baseline is the RMSE of predicting the training mean for all test points. Bootstrap resampling (10 000 iterations; Efron 1979) and permutation tests are used throughout to control for overfitting and to assess statistical significance.

3 RESULTS

3.1 The structural core

Systematic variable search identified three robustly predictive axes (Table 2). Together, the three-axis core achieves LOO gap = 44.1 per cent ($N = 45$). All three axes survive nested cross-validation (45/45 folds), bootstrap sign stability (max flip < 3 per cent), and mutual partial correlation tests.

3.2 The fourth axis: V_{flat} and VfResid

Adding $\log V_{\mathrm{flat}}$ to the core raises LOO gap to 52.1 per cent. Physical decomposition reveals that baryonic structure ($\log M_{\mathrm{bar}}$, $\log L_{3.6}$, $\log R_{\mathrm{disk}}$, morphological type T) explains 92.2 per cent of V_{flat} variance. The structural component adds only +0.4 pp above the core, while the 7.8 per cent kinematic residual (VfResid = $\log V_{\mathrm{flat}}$ − structural prediction) adds +17.1 pp.

VfResid is identified as a baryon–halo coupling proxy:

- Correlates strongly with dark halo strength (haloK, $r = +0.60$)
- Anti-correlates with MOND improvement ($r = -0.54$)
- Clean on measurement systematics (inclination, distance, quality: all $|r| < 0.08$)

In head-to-head competition, VfResid dominates every tested halo proxy by at least 7 pp. Mediation analysis shows VfResid absorbs 100 per cent of haloK's signal (Sobel-like mediation = 107 per cent), while lhOuter contributes independently.

3.3 The fifth axis: lhOuter

The outer halo fitting quality metric adds a genuine +4.3 pp above the four-axis model:

3.4 Regime dependence

The VfResid coupling signal is strongly regime-dependent (Table 4). A gradual transition occurs above $V_{\mathrm{flat}} \sim 120\ \mathrm{km\,s^{-1}}$, reaching full strength above $\sim$180 $\mathrm{km\,s^{-1}}$. In the low-V_{flat} regime ($N = 10$), the signal is completely unstable ($P(\Delta > 0) = 18.6$ per cent).

Table 5. External validation: frozen-coefficient prediction.

Regime	N	Core+VfResid gap	$r(\mathrm{VfResid}, a_0)$
Full sample	59	+8.2 per cent	0.713
$V_{\mathrm{flat}} \geq 120$ km s^{-1}	16	+34.3 per cent	0.841
$V_{\mathrm{flat}} \geq 180$ km s^{-1}	8	+48.7 per cent	0.858
Q= 1 + $V_{\mathrm{flat}} \geq 120$	11	+59.0 per cent	0.830
$V_{\mathrm{flat}} < 120$ km s^{-1}	43	−3.9 per cent	—

Table 6. Effect of improved $\log M_{\mathrm{host}}$ estimation on external performance.

Regime	Core (crude)	Core (improved)	VfResid margin
Full ($N = 59$)	−53.0 per cent	−8.1 per cent	+45.0 pp
High-V ($N = 16$)	−6.7 per cent	+16.7 per cent	+30.0 pp
Very-high-V ($N = 8$)	−49.4 per cent	−16.7 per cent	+76.3 pp
Q= 1+HV ($N = 11$)	+1.9 per cent	+25.4 per cent	+43.0 pp

3.5 Irreducible residual

What drives VfResid itself? The best five-predictor model (haloK + lhOuter + environment + MR + concentration) achieves LOO $R^2 = 0.493$, leaving 35.6 per cent of VfResid variance unexplained. This irreducible portion still adds +6.8 pp to the core model, indicating genuine dynamical physics beyond current catalogue observables.

3.6 External validation

To test whether the coupling law generalizes beyond the training sample, we assembled per-galaxy data for $N = 59$ SPARC galaxies not included in Stage A. Per-galaxy a_0 values were obtained from RAR fits; VfResid was computed using the frozen $N = 45$ structural model; and $\log M_{\mathrm{host}}$ was estimated from a V_{flat}-based formula. No refitting was performed — all predictions use frozen $N = 45$ coefficients.

3.7 Hierarchy replication and environmental decomposition

The complete hierarchical structure replicates externally. All eight hierarchy checks pass: Core alone fails in every regime, VfResid dominates in every regime (bootstrap $P > 99.6$ per cent), and lhOuter provides genuine secondary improvement. Channel dominance testing confirms VfResid outperforms every alternative proxy ($\log V_{\mathrm{flat}}$, $\log M_{\mathrm{HI}}$, $\log M_{\mathrm{bar}}$, $\log M_{\mathrm{host}}$, logMR) by margins of +35 to +71 pp.

The crude V_{flat}-based $\log M_{\mathrm{host}}$ formula correlates only $r = 0.232$ with real tidal-analysis values on the $N = 45$ training sample. We trained an improved estimator (Model C: $\log V_{\mathrm{flat}} + \log M_{\mathrm{bar}} + \log R_{\mathrm{disk}} + T$; $r = 0.542$) and repeated the external analysis. Replacing the crude formula substantially improves Core performance (Table 6), proving the environmental axis carries real signal. However, VfResid retains clear dominance (margin 30–76 pp) in every regime, with bootstrap $P = 100$ per cent. This establishes that VfResid dominance is independent of environmental proxy quality.

Table 7. Regime dependence of the hidden residual (after removing haloK).

Sample	Low-V r	High-V r	Very-high-V r
Internal ($N = 45$)	0.299	0.749	0.832
External ($N = 59$)	0.479	0.889	0.854
Pooled ($N = 104$)	0.467	0.815	0.852

3.8 Sample salvage

To maximize statistical power for the physical interpretation programme, we recovered 35 additional galaxies from 60 unused SPARC galaxies via two strategies: (a) estimating V_{flat} from the maximum observed rotation curve velocity for 22 galaxies lacking catalogue V_{flat}, and (b) lowering the minimum RAR data point threshold from 5 to 3 for 13 galaxies with sufficient but sparse data. The augmented sample ($N = 94$) preserves all previously established correlations (all differences < 0.02 in r).

3.9 VfResid driver analysis

We systematically modelled VfResid itself at three levels: internal-only ($N = 45$), external-only ($N = 94$), and pooled with source controls ($N = 139$).

haloK (dark halo linear amplitude) is the strongest predictor of VfResid in both samples (internal $r = 0.598$, external $r = 0.511$). Pure structural models fail entirely (LOO $R^2 = -0.144$ internal, 0.077 external), confirming VfResid carries genuine halo information. A large irreducible fraction persists beyond all catalogue observables: 35.8 per cent internal and 68.5 per cent external. This irreducible component still correlates with a_0 ($r = 0.438$ internal, 0.556 external), demonstrating that VfResid encodes genuine physical information beyond any available catalogue observable.

3.10 Regime law

Three tests address whether the hidden physics inside VfResid genuinely activates with V_{flat} or only becomes more observable.

After removing haloK from VfResid, the unexplained remainder (hkResid) still predicts a_0 — and this predictive power rises sharply with V_{flat} (Table 7). In the pooled sample above $V_{\mathrm{flat}} \sim 100$ km s^{-1}, the unexplained residual surpasses haloK itself in a_0-predictive power.

3.11 Physical interpretation

We test four candidate hypotheses for the hidden physics inside VfResid:

(i) **H1: Halo response** — adiabatic contraction/baryon–halo coupling (Blumenthal et al. 1986; Di Cintio et al. 2014). Proxies: baryonic surface density, disc dominance.

(ii) **H2: Assembly history** — formation time/halo age. Proxies: morphological type, surface brightness.

(iii) **H3: Feedback imprint** — SN/AGN reshaping of inner halo (Di Cintio et al. 2014; Katz et al. 2017). Proxies: gas fraction, MOND improvement.

(iv) **H4: Dynamical integration** — accumulated baryon–halo processing. Proxies: dynamical time, RC extent, RAR mean run length.

Both the slope and correlation of the hkResid–a_0 relationship strengthen with $V_{\rm flat}$ (internal: slope 1.57 at low-V to 4.13 at high-V; r from 0.201 to 0.713), ruling out a pure observability explanation.

Scoring each hypothesis by supporting evidence yields H4 (dynamical integration, score 10) as the clear leader, followed by H1 (halo response, score 7) and H3 (feedback imprint, score 7), with H2 (assembly history, score 4) weakest. The key discriminant is that logMR (dynamical coherence) is the only proxy that *regime-strengthens* in hkResid (low-V $r = 0.11 \rightarrow$ high-V $r = 0.508$), while gas fraction *fades* at high-V ($r = 0.597 \rightarrow 0.077$) — opposite the activation pattern.

The full model (all proxies) explains 34.8 per cent of hkResid (LOO $R^2 = 0.046$), reducing $r(\text{remaining}, a_0)$ from 0.651 to 0.288. Even after removing all hypothesis-specific content, the remaining signal still predicts a_0.

4 DISCUSSION

4.1 The coupling law

The empirical law can be stated as follows. Per-galaxy a_0 is governed by a hierarchical, regime-dependent baryon–halo coupling law with three tiers:

(i) **Structural core** ($M_{\rm HI}$, $M_{\rm host}$, MeanRun): sets the baseline a_0 based on gas content, halo environment, and dynamical regularity. Carries real signal (confirmed by environmental decomposition) but fails to transfer alone.

(ii) **Dominant kinematic coupling channel** (VfResid): the residual flat velocity beyond baryonic structural prediction. Carries the bulk of the halo-to-a_0 coupling signal. Mediates all tested halo proxy effects. Activates gradually above $V_{\rm flat} \sim 120\,{\rm km\,s^{-1}}$, reaching full strength above $\sim$180 km s^{-1}. Confirmed as the primary transferable channel: dominates by 30–76 pp even after improving Core environmental estimates.

(iii) **Secondary outer-halo channel** (lhOuter): independent contribution from outer rotation curve halo fitting quality. Adds $\sim$4 pp internally; +8.3 pp in very-high-$V_{\rm flat}$ externally (after correcting for $M_{\rm host}$ error absorption).

4.2 Physical interpretation

The dominant channel VfResid is not reducible to any single catalogue observable. After removing haloK, a large irreducible fraction persists (35.8 per cent internal, 68.5 per cent external) that still predicts a_0. This hidden residual activates with $V_{\rm flat}$ — both slope and correlation strengthen (slope ratio $\sim$2:1) — and hypothesis testing identifies dynamical integration (H4) as the best-supported explanation.

Gas fraction and morphology are relevant at low $V_{\rm flat}$ but fade in the high-$V_{\rm flat}$ regime where the coupling law is strongest, consistent with a transition from feedback-dominated (low mass) to integration-dominated (high mass) physics (Di Cintio et al. 2014; Katz et al. 2017). The coupling law thus encodes something deeper than simple halo strength — it reflects a regime-dependent dynamical integration effect with haloK as the best partial observable handle.

4.3 Relation to previous work

Our finding of structured a_0 variation is consistent with, but goes beyond, Rodrigues et al. (2018) and Li et al. (2018), who showed that per-galaxy RAR fits reveal a_0 variation exceeding observational uncertainties. We extend their results in three ways: (i) we identify *which* galaxy properties predict the variation, (ii) we demonstrate that the predictive hierarchy transfers to independent galaxies, and (iii) we provide a physical interpretation favouring dynamical integration.

The regime dependence of the coupling law connects to the diversity problem (Oman et al. 2015; Santos-Santos et al. 2020): the same $V_{\rm flat}$ threshold below which our law fails ($\sim$120 km s^{-1}) is the regime where halo diversity is greatest. This suggests that the low-mass failure may reflect genuine physics (incomplete baryon–halo coupling) rather than merely data quality limitations.

The environmental decomposition (Section 3.7) directly addresses the concern raised by Desmond (2017) that environmental effects may drive apparent scaling relation residuals. We show that improving the environmental proxy strengthens the Core but does not diminish VfResid dominance, confirming that the kinematic residual channel is independent of environmental estimation quality.

4.4 Implications

If a_0 variation is real and follows the coupling law described here, it has implications for both dark matter and modified gravity models:

- In ΛCDM, the regime-dependent coupling points to mass-dependent baryon–halo interaction physics (Di Cintio et al. 2014; Katz et al. 2017) that should be reproducible in hydrodynamical simulations with sufficient resolution.
- In MOND-like frameworks (Milgrom 1983; Famaey & McGaugh 2012), systematic a_0 variation would require either an environmental modification of the acceleration scale or an additional degree of freedom beyond the simple one-parameter formulation.
- The dynamical integration interpretation (H4) predicts that galaxies with longer dynamical histories should show stronger coupling — testable with resolved kinematic data from IFU surveys.

5 LIMITATIONS AND CAVEATS

(i) **Single survey**: Both training and external samples come from SPARC. Cross-survey replication (different photometry, rotation curve reduction) would substantially strengthen the claim.

(ii) **External a_0 quality**: Per-galaxy a_0 for the external sample was derived from fewer RAR data points (median 8 vs $\sim$20–30 for the training sample).

(iii) **$\log M_{\rm host}$ estimation**: Even the improved estimator (Model C, $r = 0.542$) explains only $\sim$29 per cent of real $\log M_{\rm host}$ variance. True group catalogue data would provide a definitive test.

(iv) **Regime dependence**: The coupling law works well only in the high-$V_{\rm flat}$ regime. Evidence suggests genuine physical amplification (both slope and r strengthen), but low-$V_{\rm flat}$ failure could also partly reflect data quality limitations.

(v) VfResid **construction**: The structural model used to compute VfResid is trained on the $N = 45$ sample. Applying it to external galaxies assumes the same structural relationship holds.

(vi) **Sample size**: The best-performing external subsamples have limited N (very-high-$V_{\rm flat}$: $N = 8$, Q= 1+HV: $N = 11$). Bootstrap stability is strong ($P > 99.6$ per cent), but larger samples would be valuable.

(vii) **Hypothesis testing**: The scoring is heuristic, not formal Bayesian model comparison. Hypotheses are not mutually exclusive — dynamical integration, halo response, and feedback may all contribute simultaneously.

(viii) **No claimed universality**: We report an empirical regularity within SPARC data, supported by comprehensive external validation within the same survey. Whether this reflects fundamental physics requires multi-survey validation.

6 CONCLUSIONS

Within the SPARC galaxy sample, per-galaxy a_0 variation is well-described by a hierarchical coupling law in which baryonic structure sets the baseline and a kinematic residual channel (VfResid) carries the dominant halo coupling signal. Our key findings are:

(i) The complete hierarchy replicates in $N = 94$ independent galaxies: Core alone fails, VfResid dominates by 30–76 pp, and all hierarchy checks pass (bootstrap $P > 99.6$ per cent).

(ii) VfResid is the primary transferable channel: it alone achieves 47.5 per cent gap (full) and 75.4 per cent (very-high-$V_{\rm flat}$), outperforming every alternative proxy by 35–71 pp.

(iii) VfResid dominance is independent of environmental proxy quality: improving $\log M_{\rm host}$ estimation strengthens the Core but does not diminish VfResid's margin.

(iv) Pure structural models fail to predict VfResid (LOO $R^2 < 0$), but haloK is the shared leading driver ($r = 0.60$ internal, 0.51 external). An irreducible fraction (36–69 per cent) persists and still predicts a_0.

(v) The hidden physics genuinely activates with $V_{\rm flat}$: after removing haloK, the unexplained residual predicts a_0 at $r = 0.30$ (low-V) versus $r = 0.75$ (high-V), with both slope and correlation strengthening.

(vi) Dynamical integration is the best-supported physical interpretation (score 10 vs 7/7/4), consistent with a transition from feedback-dominated to integration-dominated physics at high $V_{\rm flat}$.

The physical driver of a_0 variation is not baryonic structure per se, nor simple halo amplitude, but a regime-dependent dynamical integration effect — the accumulated product of baryon–halo interaction that genuinely amplifies in deeper potential wells.

The natural next step is testing whether this picture survives in an entirely independent survey with higher-quality data, particularly for spatially resolved kinematics that could directly probe the dynamical integration hypothesis.

ACKNOWLEDGEMENTS

We thank the SPARC team (F. Lelli, S. S. McGaugh, J. M. Schombert) for making their data publicly available.

DATA AVAILABILITY

All analysis scripts, machine-readable results, and input data are archived at Zenodo (Concept DOI: 10.5281/zenodo.19430633) for full reproducibility. The SPARC database is available at http://astroweb.cwru.edu/SPARC/.

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